

ON MAHLER'S CONJECTURE FOR EVEN s -CONCAVE FUNCTIONS IN DIMENSIONS 1 AND 2.

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ABSTRACT. In this paper, we establish different sharp forms of Mahler's conjecture for s -concave even functions in dimensions n , for $n = 1$ and 2 , for $s > -1/n$, thus generalizing our previous results in [FN] on log-concave even functions in dimension 2 , which corresponds to the case $s = 0$. The functional volume product of an even s -concave function g is

$$\int_{\mathbb{R}^n} g(x) dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y) dy,$$

where $\mathcal{L}_s g$ is the s -polar function associated to g . The analogue of Mahler's conjecture for even s -concave functions postulates that this quantity is minimized for the indicatrix of a cube for any $s > -1/n$. In dimension $n = 1$, we prove this conjecture for all $s \in (-1, 0)$ (the case $s \geq 0$ was established by the first author and Mathieu Meyer in [FM10, page 17]). In dimension $n = 2$, we consider the case $1/s \in \mathbb{Z}$: for $s > 0$, we establish Mahler's conjecture for general s -concave even functions; for $s < 0$, we prove a sharp inequality for s -concave functions g such that g^s admits an asymptote in every direction. Notice that this set of functions is quite natural to consider, when $s < 0$, since it is the largest subset of s -concave functions stable by s -duality.

1. INTRODUCTION.

The classical Blaschke-Santaló inequality gives the following sharp relation between the volume of a centrally symmetric convex body K in $\mathbb{R}^n$ and the volume of its polar body K° :

$$P(K) = |K| |K^\circ| \leq P(B_2^n),$$

where B_2^n is the Euclidean ball of radius one, $K^\circ = \{y \in \mathbb{R}^n, \langle x, y \rangle \leq 1, \forall x \in K\}$ is the polar body of K in $\mathbb{R}^n$ and $|A|$ stands for the Lebesgue measure of a Borel subset A of $\mathbb{R}^n$.

One of the well-known problems in the field of geometry of convex bodies is an inverse form of the Blaschke-Santaló inequality, namely Mahler's conjecture. This conjecture states that the product of the volume of a centrally symmetric convex body K and the volume of its polar body K° , should be minimized by the cube, that is,

$$P(K) \geq P([-1, 1]^n) = \frac{4^n}{n!}.$$

This conjecture was stated in 1938 by Mahler [Mah2]. The inequality in dimension two was established by Mahler himself [Mah1], the 3-dimensional inequality and the equality case were proved by Iriyeh and Shibata [IS], a shorter proof can be found in [FHRZ]. The conjecture was also proved for several particular families of convex bodies like unconditional convex bodies by Saint Raymond [SR, M], zonoids by Reisner [Re, GMR], hyperplane sections of $B_p^n = \{x \in \mathbb{R}^n; \sum |x_i|^p \leq 1\}$ by Karasëv [K].

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Functional forms of Mahler's conjecture have also been studied, where the centrally symmetric convex body is replaced with an even s -concave function and the polar body is replaced with the suitable s -polar of this function. More precisely, let us recall the definition of s -concavity.

Definition 1.1. Let $s \in \mathbb{R}$. A function $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ is s -concave, if for every $\lambda \in [0, 1]$, and every $x, y \in \mathbb{R}^n$ such that $g(x)g(y) > 0$, one has

$$g((1-\lambda)x + \lambda y) \geq ((1-\lambda)g(x)^s + \lambda g(y)^s)^{\frac{1}{s}}, \quad \text{for } s \neq 0,$$

and $g((1-\lambda)x + \lambda y) \geq g(x)^{1-\lambda}g(y)^\lambda$, for $s = 0$.

Let $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be an s -concave function not identically zero. Notice that for $s > 0$, it means that g^s is concave on its support, while for $s < 0$, it means that $g^s : \mathbb{R}^n \rightarrow \mathbb{R}_+ \cup \{+\infty\}$ is convex. For s -concave functions, the s -duality takes the following form. We define, for every $y \in \mathbb{R}^n$,

$$\mathcal{L}_s g(y) := \inf_{x \in \mathbb{R}^n} \frac{(1 - s\langle x, y \rangle)_+^{\frac{1}{s}}}{g(x)}, \quad \text{for } s \neq 0, \quad \text{and} \quad \mathcal{L}_0 g(y) = \inf_{x \in \mathbb{R}^n} \frac{e^{-\langle x, y \rangle}}{g(x)},$$

where the infimum are considered on the set $\{x \in \mathbb{R}^n; g(x) > 0\}$. Note that $\mathcal{L}_s g$ is s -concave and that $\mathcal{L}_0$ can also be described using the Legendre transform: for any function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$, one has $\mathcal{L}_0(e^{-\varphi}) = e^{-\varphi^*}$, where $\varphi^*(y) = \sup_x (\langle x, y \rangle - \varphi(x))$ denotes the Legendre transform of φ .

The following conjecture is the natural analogue of Mahler's conjecture for even s -concave functions.

Conjecture 1.2. Let $s > -\frac{1}{n}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be an even s -concave function such that $0 < \int g < +\infty$. Then,

$$\int_{\mathbb{R}^n} g(x) dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y) dy \geq \frac{4^n}{(1+s) \cdots (1+ns)},$$

with equality for $g = \mathbb{1}_{[-1,1]^n}$.

The conjecture has been proved for $s = 0$ and g being unconditional in [FM08a, FM08b]. Recall that a function is unconditional if $g(|x_1|, \dots, |x_n|) = g(x_1, \dots, x_n)$ for every $(x_1, \dots, x_n) \in \mathbb{R}^n$. It has also been established for $s > 0$ and $n = 1$ in [FM10, page 17]; for $s > 0$ and g unconditional in [BF, FGSZ] and for $s = 0$ and $n = 2$ in [FN].

According to the sign of s , there are two ways of reformulating Conjecture 1.2. If $s > 0$, then it is equivalent to finding the exact lower bound for the quantity

$$\int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}_1 f(y))^m dy,$$

where f is an even concave function such that $0 < \int f^m < +\infty$. While for $s < 0$, Conjecture 1.2 is equivalent to finding the exact lower bound for

$$\int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy,$$

where $f : \mathbb{R}^n \rightarrow (0, +\infty]$ is even, convex, f goes to infinity at infinity and for every $y \in \mathbb{R}^n$

$$\mathcal{M}f(y) = \sup_{x \in \mathbb{R}^n} \frac{1 + \langle x, y \rangle}{f(x)}.$$

In our first main result, we prove Conjecture 1.2 in the case where $1/s \in \mathbb{N}$, and in dimension $n = 2$.

Theorem 1.3. Let $s > 0$ be such that $1/s \in \mathbb{N}$ and let $g : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be an even s -concave function such that $0 < \int g < +\infty$. Then,

$$\int_{\mathbb{R}^2} g(x) dx \int_{\mathbb{R}^2} \mathcal{L}_s g(y) dy \geq \frac{16}{(1+s)(1+2s)},$$

with equality for $g = \mathbb{1}_{[-1,1]^2}$.

Notice that letting $s \rightarrow 0$, we get a new proof of the log-concave case from [FN], though without the equality case, see Corollary 3.3.

For $s < 0$, the situation is more involved, because the s -dual $\mathcal{L}_s g$ of an integrable s -concave function g is not only s -concave but, as observed in [Ro, FGSZ], the function $(\mathcal{L}_s g)^s$ has also asymptotes in every direction. For example, for $g = \mathbb{1}_{[-1,1]^n}$, one has $(\mathcal{L}_s g(y))^s = 1 - s\|y\|_1$, where $\|y\|_1 = \sum_{i=1}^n |y_i|$, for $y = (y_1, \dots, y_n)$. More precisely, in general, one has $(\mathcal{L}_s g)^s \in \mathcal{F}$, where the set $\mathcal{F}$ is defined below.

Definition 1.4. $\mathcal{F}$ is the set of convex lower semi-continuous functions $f : \mathbb{R}^n \rightarrow (0, +\infty)$ such that, for any $x \neq 0$, one has $\lim_{t \rightarrow +\infty} f(tx) = +\infty$, and $t \mapsto \frac{f(tx)}{t}$ is non-increasing on $(0, +\infty)$.

Notice that if $g^s \in \mathcal{F}$, then it is not difficult to see that $0 < \int g < +\infty$, see the inequality (2.3) in section 2. Moreover, it is easy to see that for any g such that $g^s \in \mathcal{F}$, one has $\mathcal{L}_s \mathcal{L}_s g = g$, from which it follows that for any map g , one has $\mathcal{L}_s \mathcal{L}_s \mathcal{L}_s g = \mathcal{L}_s g$. As we shall see from the one-dimensional case, it is natural to formulate two more conjectures from which Conjecture 1.2 follows in the case $s < 0$. The first one postulates a sharp lower bound on the functional volume product of s -concave even functions, among the ones such that $g^s \in \mathcal{F}$.

Conjecture 1.5. Let $s \in (-\frac{1}{n}, 0)$ and let $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be even such that $g^s \in \mathcal{F}$. Then,

$$\int_{\mathbb{R}^n} g(x) dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y) dy \geq \frac{1}{(1+ns)} \times \frac{4^n}{(1+s) \cdots (1+ns)},$$

with equality for $g(x) = (\max(1, \|x\|_\infty))^{1/s}$.

Notice that $(\max(1, \|x\|_\infty))^{1/s} = \mathcal{L}_s \mathcal{L}_s (\mathbb{1}_{[-1,1]^n})$. In [FGSZ], the above conjecture was established if the function g is unconditional. In particular, the conjecture holds for $n = 1$. In our second main result, we establish Conjecture 1.5 when $n = 2$ and $-1/s \in \mathbb{N}$.

Theorem 1.6. Let $s \in (-\frac{1}{2}, 0)$ be such that $-1/s \in \mathbb{N}$ and let $g : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be even such that $g^s \in \mathcal{F}$. Then,

$$\int_{\mathbb{R}^2} g(x) dx \int_{\mathbb{R}^2} \mathcal{L}_s g(y) dy \geq \frac{16}{(1+s)(1+2s)^2},$$

with equality for $g(x) = (\max(1, \|x\|_\infty))^{1/s}$.

The second conjecture compares the integral of g with the integral of its double s -polar.

Conjecture 1.7. Let $s \in (-\frac{1}{n}, 0)$ and let $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be even s -concave such that $0 < \int g < +\infty$. Then,

$$\int_{\mathbb{R}^n} g(x) dx \geq (1+ns) \int_{\mathbb{R}^n} \mathcal{L}_s \mathcal{L}_s g(x) dx, \quad (1.1)$$

with equality for $g = \mathbb{1}_{[-1,1]^n}$.

Notice that, for any function $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$, one has, for any $x \in \mathbb{R}^n$, $g(x) \leq \mathcal{L}_s \mathcal{L}_s g(x)$ thus $\int_{\mathbb{R}^n} g(x) dx \leq \int_{\mathbb{R}^n} \mathcal{L}_s \mathcal{L}_s g(x) dx$. The conjectured inequality (1.1) postulates that one has an opposite bound up to constant for s -concave even functions.

In our third main result, we establish Conjecture 1.7 in dimension $n = 1$.

Theorem 1.8. Let $s \in (-1, 0)$ and let $g : \mathbb{R} \rightarrow \mathbb{R}_+$ be even s -concave such that $0 < \int g < +\infty$. Then,

$$\int_{\mathbb{R}} g(x) dx \geq (1+s) \int_{\mathbb{R}} \mathcal{L}_s \mathcal{L}_s g(y) dy,$$

with equality for $g = \mathbb{1}_{[-1,1]}$.

Remark 1.9. Notice that, if Conjecture 1.7 holds for any s -concave even function g and Conjecture 1.5 holds for $\mathcal{L}_s g$, then multiplying equation (1.1) by $\int \mathcal{L}_s g$, using that $(\mathcal{L}_s g)^s \in \mathcal{F}$ and applying Conjecture 1.5 to $\mathcal{L}_s g$, then we deduce that Conjecture 1.2 holds for g .

From the above remark, the results in [FGSZ] and Theorem 1.8, one deduces that Conjecture 1.2 holds for $n = 1$ and $s \in (-1, 0)$, which is the following theorem.

Theorem 1.10. *Let $s \in (-1, 0)$ and $g : \mathbb{R} \rightarrow \mathbb{R}_+$ be even convex such that $0 < \int g < +\infty$. Then,*

$$\int_{\mathbb{R}} g(x) dx \int_{\mathbb{R}} \mathcal{L}_s g(y) dy \geq \frac{4}{(1+s)^2},$$

with equality if and only if $g = c\mathbb{1}_{[-a,a]}$, for some $c > 0$ and $a > 0$.

The proof's strategy of our main theorems in dimension $n = 2$ relies on constructing an associated symmetric convex body in dimension $m + 2$, where $m = 1/|s|$, to which we apply the following proposition concerning Mahler's conjecture in higher dimensions proved in [FHMRZ].

Proposition 1.11. [FHMRZ] *Let $K \subset \mathbb{R}^n$ be a centrally symmetric convex body that can be partitioned with hyperplanes $H_1, H_2, \dots, H_n$ into 2^n pieces of the same volume such that each section $K \cap H_i$ satisfies Mahler's conjecture and is partitioned into 2^{n-1} regions of the same $(n-1)$ -dimensional volume by the remaining hyperplanes, then*

$$P(K) \geq \frac{4^n}{n!}.$$

In our proofs, we shall need the following notion of equipartition of functions.

Definition 1.12. Let $m \in \mathbb{Z}$. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}_+$ is said to be m -equipartitioned if

$$\int_{\mathbb{R}_{\varepsilon}^n} f^m = \frac{1}{2^n} \int_{\mathbb{R}^n} f^m \quad \text{and} \quad \int_{\mathbb{R}_{\varepsilon}^n} f^{m-1} = \frac{1}{2^n} \int_{\mathbb{R}^n} f^{m-1},$$

where $\forall \varepsilon \in \{-1, 1\}^n$, $\mathbb{R}_{\varepsilon}^n = \{x \in \mathbb{R}^n; \varepsilon_i x_i \geq 0, \forall i \in \{1, \dots, n\}\}$.

This paper is organized in the following way: In section 2, we present some general results and properties of the s -dual function $\mathcal{L}_s$ and we introduce the construction of convex bodies using s -concave functions. In section 3, we establish Conjecture 1.2 in dimensions $n = 1$ and $n = 2$ for $\frac{1}{s} \in \mathbb{N}$. In section 4, we give a lower bound of the functional volume product applied to even s -concave functions in dimensions $n = 1$ and $n = 2$ for $-\frac{1}{s} \in \mathbb{N}$.

2. CONSTRUCTION AND PROPERTIES OF ASSOCIATED CONVEX BODIES.

Let $s > -\frac{1}{n}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be an s -concave function. We discuss two cases according to the sign of s .

2.1. Case $s > 0$. Let $m = \frac{1}{s} > 0$. Assume that $0 < \int g < +\infty$. Then there exists a function $f : \mathbb{R}^n \rightarrow \mathbb{R}_+$ concave on its support such that $g = f^{\frac{1}{s}} = f^m$ and

$$\mathcal{L}_s g(y) = \mathcal{L}_s f^{\frac{1}{s}}(y) = \inf_{x \in \mathbb{R}^n} \frac{(1 - s\langle x, y \rangle)_+^{\frac{1}{s}}}{f(x)^{\frac{1}{s}}} = \left(\inf_{x \in \mathbb{R}^n} \frac{(1 - \langle x, sy \rangle)_+}{f(x)} \right)^{\frac{1}{s}} = (\mathcal{L}f(sy))^{\frac{1}{s}} = \left(\mathcal{L}f\left(\frac{y}{m}\right) \right)^m,$$

where $\mathcal{L}f(y) = \mathcal{L}_1 f(y) = \inf_{x \in \mathbb{R}^n} \frac{(1 - \langle x, y \rangle)_+}{f(x)}$. In addition, by the change of variables $y = mz$, we get

$$\int_{\mathbb{R}^n} \mathcal{L}_s g(y) dy = m^n \int_{\mathbb{R}^n} (\mathcal{L}f(z))^m dz.$$

Introducing the operator P_m defined as follows

$$P_m(f) = \int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy,$$

and changing variables, we get

$$\int_{\mathbb{R}^n} g(x) dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y) dy = m^n P_m(f).$$

Thus, for $s > 0$, Conjecture 1.2 can equivalently be formulated in the following form: for any even function $f : \mathbb{R}^n \rightarrow \mathbb{R}_+$, concave on its support and integrable and for any $m > 0$, one should have

$$P_m(f) = \int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy \geq P_m(\mathbf{1}_{[-1,1]^n}) = \frac{4^n}{(m+1) \cdots (m+n)}. \quad (2.1)$$

The operator P_m is invariant under linear transformation. Indeed, let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an invertible linear map. Then, putting $z = Tx$ we get,

$$\int_{\mathbb{R}^n} (f \circ T(x))^m dx = \frac{1}{|\det T|} \int_{\mathbb{R}^n} f(z)^m dz.$$

We also obtain, for every $y \in \mathbb{R}^n$,

$$\mathcal{L}(f \circ T)(y) = \inf_{x \in \mathbb{R}^n} \frac{(1 - \langle x, y \rangle)_+}{f(Tx)} = \inf_{z \in \mathbb{R}^n} \frac{(1 - \langle T^{-1}z, y \rangle)_+}{f(z)} = \mathcal{L}f((T^{-1})^*y).$$

Therefore, $\mathcal{L}(f \circ T) = (\mathcal{L}f) \circ (T^{-1})^*$. Hence, changing variables, we get

$$\begin{aligned} P_m(f \circ T) &= \int_{\mathbb{R}^n} (f(Tx))^m dx \int_{\mathbb{R}^n} (\mathcal{L}f((T^{-1})^*y))^m dy \\ &= \frac{1}{|\det T|} \int_{\mathbb{R}^n} f(x)^m dx \times |\det T^*| \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy = P_m(f). \end{aligned}$$

Let us introduce some tools that will help us in the proof of our main theorem in section 3.

Let L be a symmetric convex body in $\mathbb{R}^m$. Define the set

$$K_{L,m}(f) = \{(x, t) \in \mathbb{R}^n \times \mathbb{R}^m; \|t\|_L \leq f(x)\},$$

where $\|\cdot\|_L$ is defined for $y \in \mathbb{R}^m$ by $\|y\|_L = \inf\{\lambda > 0; y \in \lambda L\}$. Variants of these sets and their volume products were considered in [AKM, BF] but we establish their properties below for completeness. Since f is even and concave, then $K_{L,m}(f)$ is a symmetric convex body in $\mathbb{R}^{m+n}$ and for every $m > 0$ one has

$$|K_{L,m}(f)| = |\{(x, t) \in \mathbb{R}^n \times \mathbb{R}^m; \|t\|_L \leq f(x)\}| = \int_{\mathbb{R}^n} |f(x)L|_m dx = |L| \int_{\mathbb{R}^n} f(x)^m dx.$$

Moreover, we have

$$\begin{aligned} (K_{L,m}(f))^\circ &= \{(y, s) \in \mathbb{R}^n \times \mathbb{R}^m; \langle x, y \rangle + \langle s, t \rangle \leq 1, \quad \forall (x, t) \in K_{L,m}(f)\} \\ &= \{(y, s) \in \mathbb{R}^n \times \mathbb{R}^m; \sup_{\{t \in \mathbb{R}^m; \|t\|_L \leq f(x)\}} \langle s, t \rangle \leq 1 - \langle x, y \rangle, \quad \forall x \in \mathbb{R}^n\} \\ &= \{(y, s) \in \mathbb{R}^n \times \mathbb{R}^m; f(x) \|s\|_{L^\circ} \leq (1 - \langle x, y \rangle)_+, \quad \forall x \in \mathbb{R}^n\} \\ &= \{(y, s) \in \mathbb{R}^n \times \mathbb{R}^m; \|s\|_{L^\circ} \leq \mathcal{L}f(y)\} \\ &= K_{L^\circ, m}(\mathcal{L}f). \end{aligned}$$

Hence

$$|K_{L,m}(f)^\circ| = |K_{L^\circ, m}(\mathcal{L}f)| = |L^\circ| \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy.$$

Thus, one has

$$P(K_{L,m}(f)) = |K_{L,m}(f)| |K_{L^\circ, m}(\mathcal{L}f)| = |L| |L^\circ| \int_{\mathbb{R}^n} f(x)^m dx \int_{\mathbb{R}^n} (\mathcal{L}f(y))^m dy = P(L) P_m(f).$$

We need the following lemma describing the relation between the concave function f and its dual.

Lemma 2.1. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be a concave function on its support. Then,*

$$\text{supp}(\mathcal{L}f) = (\text{supp}f)^\circ.$$

Proof. Let $C = \text{supp}(f)$ and $K(f) = \{(x, t) \in C \times \mathbb{R}; |t| \leq f(x)\}$. We know that $K(f)$ is convex in $\mathbb{R}^{n+1}$ and $K(f)^\circ = K(\mathcal{L}f)$. Since the operation of duality transforms sections of a given convex body into projections of its polar and

$$K(f) \cap e_{n+1}^\perp = C = P_{e_{n+1}^\perp}(K(f)),$$

we have

$$K(\mathcal{L}f) \cap e_{n+1}^\perp = K(f)^\circ \cap e_{n+1}^\perp = \left(P_{e_{n+1}^\perp}(K(f))\right)^\circ.$$

Thus,

$$\text{supp}\mathcal{L}f = C^\circ = (\text{supp}f)^\circ.$$

□

2.2. Case $-\frac{1}{n} < s < 0$. Let $m = -\frac{1}{s} > n$. Then there exists a convex function $f : \mathbb{R}^n \rightarrow (0, +\infty]$ such that $f = g^s$. Assume that $0 < \int g < +\infty$, one has $\lim_{t \rightarrow +\infty} f(tx) = +\infty$, for any $x \neq 0$. Moreover, since $s = -1/m$ and $g = f^{-m}$,

$$\mathcal{L}_s g(y) = \inf_{x \in \mathbb{R}^n} \frac{(1 - s\langle x, y \rangle)_+^{\frac{1}{s}}}{g(x)} = \left(\inf_{x \in \mathbb{R}^n} \frac{(1 + \langle x, \frac{y}{m} \rangle)_+^{-1}}{f^{-1}(x)} \right)^m = \left(\mathcal{L}_{-1}(f^{-1})\left(\frac{y}{m}\right) \right)^m.$$

Let us define the function $\mathcal{M}f$, for $y \in \mathbb{R}^n$, by

$$\mathcal{M}f(y) = (\mathcal{L}_{-1}(f^{-1})(y))^{-1} = \sup_{x \in \mathbb{R}^n} \frac{1 + \langle x, y \rangle}{f(x)}.$$

It was proved in [FGSZ] that, with these hypotheses, the function $\mathcal{M}f$ belongs to $\mathcal{F}$ (see definition 1.4). Using the change of variables $y = mz$, we get

$$\int_{\mathbb{R}^n} \mathcal{L}_s g(y) dy = m^n \int_{\mathbb{R}^n} (\mathcal{L}_{-1}(f^{-1})(z))^m dz = m^n \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(z))^m} dz.$$

Consider the operator Q_m defined as follows

$$Q_m(f) = \int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy,$$

one has

$$\int_{\mathbb{R}^n} g(x) dx \int_{\mathbb{R}^n} \mathcal{L}_s g(y) dy = m^n \int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(z))^m} dz = m^n Q_m(f).$$

Thus, for $s < 0$, Conjecture 1.2 can equivalently be formulated in the following form: for any even function $f : \mathbb{R}^n \rightarrow (0, +\infty]$, convex, such that $\lim_{t \rightarrow +\infty} f(tx) = +\infty$, for any $x \neq 0$ and for any $m > n$, one should have

$$Q_m(f) = \int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy \geq Q_m(1 + I_{[-1,1]^n}) = \frac{4^n}{(m-1) \cdots (m-n)}, \quad (2.2)$$

where, for any $K \subset \mathbb{R}^n$, the function I_K is defined by $I_K(x) = 0$ for $x \in K$ and $I_K(x) = +\infty$ for $x \notin K$.

On the other hand, if $f \in \mathcal{F}$, then it was proved in [FGSZ] that for any $u \in S^{n-1}$ there exists a continuous positive function a such that $f(tu) \geq a(u)t$ for every $t > 0$. By compactness of the sphere and continuity of a there exists $c > 0$ such that $f(tu) \geq ct$ for every $t > 0$. We deduce that $f(x) \geq c\|x\|_2$ for every $x \in \mathbb{R}^n$. Moreover, since f is positive, continuous and goes

to infinity at infinity, there exists $c' > 0$ such that $f(x) \geq c'$ for every $x \in \mathbb{R}^n$. We conclude that $f(x) \geq \max(c\|x\|_2, c')$. Thus,

$$g(x) = f(x)^{\frac{1}{s}} \leq \max(c\|x\|_2, c')^{\frac{1}{s}}, \quad (2.3)$$

which is integrable since $-\frac{1}{n} < s < 0$. Therefore, for any $f \in \mathcal{F}$, the function $g = f^{\frac{1}{s}}$ is integrable.

Same as before, Conjecture 1.5 can be reformulated in the form: for $f \in \mathcal{F}$ even and $m > n$,

$$Q_m(f) \geq Q_m(\max(1, \|x\|_\infty)) = \frac{m}{m-n} \times \frac{4^n}{(m-1) \cdots (m-n)}. \quad (2.4)$$

And Conjecture 1.7 reads as: for any even function $f : \mathbb{R}^n \rightarrow (0, +\infty]$, convex such that $\lim_{t \rightarrow +\infty} f(tx) = +\infty$, for any $x \neq 0$ and for any $m > n$,

$$\int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \geq \frac{m-n}{m} \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(x))^m} dx, \quad (2.5)$$

with equality for $f = 1 + I_{[-1,1]^n}$.

Similarly as before, one may easily prove that the operator Q_m is invariant under linear transformation. Let $K \subset \mathbb{R}^m$ be a symmetric convex body. For any even convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}_+$ such that $\lim_{x \rightarrow \infty} f(tx) = +\infty$, for any $x \neq 0$, we define the function

$$\begin{aligned} \pi_m(f) : \mathbb{R}^n \times \mathbb{R}^m &\rightarrow \mathbb{R}_+ \cup \{+\infty\} \\ (x, s) &\mapsto \pi_m(f)(x, s) = \begin{cases} \|s\|_K f\left(\frac{x}{\|s\|_K}\right) & \text{if } s \neq 0. \\ \lim_{s \rightarrow 0} \|s\|_K f\left(\frac{x}{\|s\|_K}\right) & \text{if } s = 0. \end{cases} \end{aligned}$$

We denote the domain of f by $\text{dom}(f) = \{x \in \mathbb{R}^n; f(x) < +\infty\}$. Then

$$\text{dom}(\pi_m(f)) = \{(x, s) \in \mathbb{R}^n \times \mathbb{R}^m; \pi_m(f)(x, s) < +\infty\} = \{(x, s) \in \mathbb{R}^n \times \mathbb{R}^m; x \in \|s\|_K \text{dom}(f)\}.$$

Consider the following convex body

$$C_m^K(f) = \{(x, s) \in \mathbb{R}^n \times \mathbb{R}^m; \pi_m(f)(x, s) \leq 1\} = \overline{\left\{ (x, s) \in \mathbb{R}^n \times \mathbb{R}^m \setminus \{0\}; \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1 \right\}}.$$

One has,

$$\begin{aligned} (C_m^K(f))^\circ &= \{(y, t) \in \mathbb{R}^n \times \mathbb{R}^m; \langle (x, s), (y, t) \rangle \leq 1, \forall (x, s) \in C_m^K(f)\} \\ &= \left\{ (y, t) \in \mathbb{R}^n \times \mathbb{R}^m; \langle x, y \rangle + \langle s, t \rangle \leq 1, \forall (x, s); \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1 \right\}. \end{aligned}$$

Let $z = \frac{x}{\|s\|_K}$, then,

$$\begin{aligned} (C_m^K(f))^\circ &= \left\{ (y, t) \in \mathbb{R}^n \times \mathbb{R}^m; \|s\|_K \langle z, y \rangle + \langle s, t \rangle \leq 1, \forall (z, s), f(z) \leq \frac{1}{\|s\|_K} \right\} \\ &= \left\{ (y, t) \in \mathbb{R}^n \times \mathbb{R}^m; \sup_{\{s; \|s\|_K \leq 1/f(z)\}} \|s\|_K \langle z, y \rangle + \langle s, t \rangle \leq 1, \forall z \right\} \\ &= \left\{ (y, t) \in \mathbb{R}^n \times \mathbb{R}^m; \frac{\langle z, y \rangle + \|t\|_{K^\circ}}{f(z)} \leq 1, \forall z \right\} \\ &= \left\{ (y, t) \in \mathbb{R}^n \times \mathbb{R}^m; \sup_{z \in \mathbb{R}^n} \frac{\langle z, y \rangle + \|t\|_{K^\circ}}{f(z)} \leq 1 \right\} \\ &= \overline{\left\{ (y, t) \in \mathbb{R}^n \times \mathbb{R}^m \setminus \{0\}; \|t\|_{K^\circ} \mathcal{M}f\left(\frac{y}{\|t\|_{K^\circ}}\right) \leq 1 \right\}} \\ &= C_m^{K^\circ}(\mathcal{M}f). \end{aligned}$$

Let us compute the volume of the convex body $C_m^K(f)$.

$$\begin{aligned} |C_m^K(f)| &= \left| \left\{ (x, s) \in \mathbb{R}^n \times \mathbb{R}^m \setminus \{0\}; \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1 \right\} \right| \\ &= \int_{\mathbb{R}^m} \left| \left\{ x \in \mathbb{R}^n; \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1 \right\} \right|_n ds \\ &= \int_{\mathbb{R}^m} \|s\|_K^n \left| \left\{ z \in \mathbb{R}^n; f(z) \leq \frac{1}{\|s\|_K} \right\} \right|_n ds = \int_{\mathbb{R}^m} \varphi(\|v\|_K) dv, \end{aligned}$$

where $\varphi(t) = t^n |\{z \in \mathbb{R}^n; f(z) \leq \frac{1}{t}\}|_n$. Using Fubini and a change of variables, one has

$$\int_{\mathbb{R}^m} \varphi(\|v\|_K) dv = \int_0^{+\infty} (-\varphi'(t)) |\{v \in \mathbb{R}^m; \|v\|_K \leq t\}| dt = |K| \int_0^{+\infty} t^m (-\varphi'(t)) dt = m|K| \int_0^{+\infty} t^{m-1} \varphi(t) dt.$$

Hence, we get

$$\begin{aligned} |C_m^K(f)| &= m|K| \int_0^{+\infty} t^{m-1} t^n \left| \left\{ z \in \mathbb{R}^n; f(z) \leq \frac{1}{t} \right\} \right|_n dt \\ &= m|K| \int_0^{+\infty} \frac{1}{u^{m+n+1}} |\{z \in \mathbb{R}^n; f(z) \leq u\}| du \\ &= m|K| \int_{\mathbb{R}^n} \int_{f(z)}^{+\infty} \frac{du}{u^{m+n+1}} dz \\ &= \frac{m}{m+n} |K| \int_{\mathbb{R}^n} \frac{dz}{f(z)^{m+n}}. \end{aligned}$$

Using that $C_m^K(f)^\circ = C_m^{K^\circ}(\mathcal{M}f)$, we get

$$|C_m^K(f)^\circ| = \frac{m}{m+n} |K^\circ| \int_{\mathbb{R}^n} \frac{dz}{(\mathcal{M}f(z))^{m+n}}.$$

Thus, we obtain for any $m > 0$

$$\int_{\mathbb{R}^n} \frac{1}{f(x)^{m+n}} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^{m+n}} dy = \left(\frac{m+n}{m} \right)^2 \frac{1}{|K||K^\circ|} |C_m^K(f)| |C_m^K(f)^\circ|. \quad (2.6)$$

3. PROOF OF THE INEQUALITIES IN DIMENSION 1 AND 2 FOR $s > 0$.

In this section, we start by giving a monotonicity result, which implies Conjecture 1.2 in dimension 1 where $\frac{1}{s} \in \mathbb{N}$. Recall that Conjecture 1.2 in dimension 1 was proved in [FM10, page 17] for any $s > 0$. We present here a simpler proof in the case where $\frac{1}{s} \in \mathbb{N}$. Then, we present a proof of Theorem 1.3, which is Mahler's conjecture for s -concave functions in the case $n = 2$ and $\frac{1}{s} \in \mathbb{N}$.

Let $m = \frac{1}{s}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be an even s -concave function, then there exists an even function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, concave on its support, such that $g = f^m$. Thus, in dimension $n = 1$ and for $1/s \in \mathbb{N}$, Conjecture 1.2 is a consequence of the following theorem.

Theorem 3.1. *Let $m \in \mathbb{N}$, $a > 0$, $f : \mathbb{R} \rightarrow \mathbb{R}_+$ be an even concave function on its support $C = [-a, a]$ and*

$$P_m(f) = \int_{\mathbb{R}} f(x)^m dx \int_{\mathbb{R}} (\mathcal{L}f(y))^m dy.$$

Then the sequence $((m+1)^2 P_m(f) - 4m)_m$ is non-decreasing. Hence, one has $P_m(f) \geq \frac{4}{m+1}$. Moreover, if $P_m(f) = \frac{4}{m+1}$ then either f is constant on its support $[-a, a]$ or there is $c > 0$ such that $f(x) = c \left(1 - \frac{|x|}{a}\right)_+$.

Proof. First, using Lemma 2.1, we know that $\text{supp}(\mathcal{L}f) = C^\circ = [-\frac{1}{a}, \frac{1}{a}]$. Set

$$I_m = (m+1) \int_0^a f(x)^m dx \quad \text{and} \quad J_m = (m+1) \int_0^{\frac{1}{a}} (\mathcal{L}f(y))^m dy.$$

The monotonicity that we want to prove reduces to $I_m J_m \geq m+1$, for every $m \in \mathbb{N}$. The inequality holds for $m = 0$ (there is equality). Let $m \in \mathbb{N}^*$. Being concave on $[-a, a]$, the function f is left differentiable on $(-a, a]$. By concavity, denoting by $f'(t)$ the left derivative of f at t , we have for all $t \in (-a, a]$ and $x \in [-a, a]$

$$f(x) \leq f(t) + (x-t)f'(t). \quad (3.1)$$

multiplying both sides of the inequality by $mf(t)^{m-1}$ and integrating on $[0, a]$ gives

$$f(x)I_{m-1} = f(x) \int_0^a mf(t)^{m-1} dt \leq \int_0^a mf(t)^m dt + \int_0^a m(x-t)f'(t)f(t)^{m-1} dt.$$

Integration by parts gives that for every $x \in [-a, a]$

$$\int_0^a m(x-t)f'(t)f(t)^{m-1} dt = (x-a)f(a)^m - xf(0)^m + \int_0^a f(t)^m dt.$$

Using that $(x-a)f(a)^m \leq 0$, we get that for all $x \in [-a, a]$,

$$f(x)I_{m-1} \leq I_m - xf(0)^m. \quad (3.2)$$

Hence, one has for all $x \in [-a, a]$

$$\frac{I_{m-1}}{I_m} \leq \frac{1}{f(x)} \left(1 - \frac{xf(0)^m}{I_m} \right).$$

By taking the infimum over all $x \in [-a, a]$, and denoting $y_0 = \frac{f(0)^m}{I_m}$ we get

$$\frac{I_{m-1}}{I_m} \leq \mathcal{L}f(y_0).$$

Since $\mathcal{L}f$ is concave, we can apply the result to $\mathcal{L}f$ and, denoting $x_0 = \frac{(\mathcal{L}f(0))^m}{J_m}$, we get

$$\frac{J_{m-1}}{J_m} \leq f(x_0).$$

On the other hand, the definition of $\mathcal{L}f$ implies that for all $x \in \text{supp}f$, $y \in \text{supp}\mathcal{L}f$

$$f(x)\mathcal{L}f(y) \leq (1-xy)_+.$$

Using (3.2) for $x = 0$ gives that $I_m \geq f(0)I_{m-1}$, thus $I_m \geq f(0)^m a$. In the same way, one has $J_m \geq \mathcal{L}f(0)^m a^{-1}$. Multiplying these inequalities and using that f is even and concave, we get $\mathcal{L}f(0) = \max(f)^{-1} = f(0)^{-1}$, we conclude that $I_m J_m \geq 1$, which implies that $x_0 y_0 \leq 1$. Hence, one has

$$\frac{I_{m-1}J_{m-1}}{I_m J_m} \leq f(x_0)\mathcal{L}f(y_0) \leq 1 - x_0 y_0 = 1 - \frac{1}{I_m J_m}. \quad (3.3)$$

Finally we get $I_{m-1}J_{m-1} \leq I_m J_m - 1$. Hence the sequence $(I_m J_m - m)_m$ is non-decreasing. Since $I_m J_m = \frac{(m+1)^2}{4} P_m(f)$, we deduce that the sequence $u_m = (m+1)^2 P_m(f) - 4m$ is non-decreasing. It follows that $u_m \geq u_0 = P_0(f) = 4$, hence $P_m(f) \geq \frac{4}{m+1}$, for any integer m .

Assume now that there is equality, for some integer m : $P_m(f) = \frac{4}{m+1}$. Then, there is equality in (3.3). It follows that there is equality in (3.2) for $x = x_0$ and also $(x_0 - a)f(a)^m = 0$. Moreover, there is also equality in the corresponding inequality for $\mathcal{L}f$ instead of f and y_0 instead of x_0 , thus $(y_0 - \frac{1}{a})\mathcal{L}f(1/a)^m = 0$. In addition, there is equality in (3.1) for $x = x_0$ and for almost any $t \in [-a, a]$:

$$f(t) = f(x_0) + (t - x_0)f'(t).$$

On the other hand, one has $f(t) = f(x_0) + \int_{x_0}^t f'(u)du$. Since the left derivative f' is non-increasing, we deduce that f' is constant on $[0, x_0]$ and on $(x_0, a]$, which means that f is affine on $[0, x_0]$ and on $(x_0, a]$. In the same way, $\mathcal{L}f$ is affine on $[0, y_0]$ and on $(y_0, 1/a]$.

If $x_0 = a$, then f is affine on $[0, a]$ and $J_m = \mathcal{L}f(0)^m/a$. But, by concavity, one has for every $y \in [0, 1/a]$

$$\mathcal{L}f(y) \geq \mathcal{L}f(0)(1 - ay).$$

Raising this inequality to the power m and integrating, one gets that $J_m \geq \mathcal{L}f(0)^m/a$. Since there is equality we deduce that $\mathcal{L}f(y) = \mathcal{L}f(0)(1 - ay)$, for every $y \in [0, 1/a]$. Thus $f = f(0)\mathbb{1}_{[-a, a]}$.

If $x_0 \neq a$, then $f(a) = 0$. Thus there exists $0 \leq b \leq 1/a$ and $c > 0$ such that, for $t \geq 0$,

$$f(t) = c(1 - bt)\mathbb{1}_{[0, x_0]} + c(1 - bx_0)\frac{a - x}{a - x_0}\mathbb{1}_{(x_0, a]}.$$

This implies that, for any $y \geq 0$,

$$\mathcal{L}f(y) = \frac{1}{c} \left(\mathbb{1}_{[0, b]}(y) + \frac{1 - x_0 y}{1 - bx_0} \mathbb{1}_{(b, \frac{1}{a}]}(y) \right).$$

Since $x_0 \neq a$, one has $\mathcal{L}f(1/a) \neq 0$, hence $y_0 = 1/a$. This gives $I_m = af(0)^m$ and this implies as before that $f(x) = f(0)(1 - \frac{x}{a})$, for all $x \in [0, a]$. □

Let us prove the 2-dimensional case of our inequality for m being an integer.

Theorem 3.2. *Let $m \in \mathbb{N}$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be an even function, concave on its support, such that $0 < \int f < +\infty$. Then,*

$$P_m(f) \geq \frac{16}{(m+1)(m+2)}.$$

Moreover, there is equality for $f = \mathbb{1}_{[-1, 1]^2}$.

As noticed before, this theorem may be reformulated in the following form: for any integer $m > 0$ and any even function $g : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ that is $\frac{1}{m}$ -concave, integrable with $\int_{\mathbb{R}^2} g > 0$ one has

$$\int_{\mathbb{R}^2} g \int_{\mathbb{R}^2} \mathcal{L}_{\frac{1}{m}} g \geq \frac{16m^2}{(m+1)(m+2)}. \quad (3.4)$$

The following corollary proves that letting m go to infinity we get a new proof of Mahler conjecture for even log-concave functions, which was our previous main result in [FN].

Corollary 3.3. *Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be an even log-concave function such that $0 < \int f < +\infty$, then*

$$\int_{\mathbb{R}^2} f(x)dx \int_{\mathbb{R}^2} f^\circ(y)dy \geq 16, \quad (3.5)$$

where $f^\circ(y) = \inf_{x \in \mathbb{R}^n} \frac{e^{-\langle x, y \rangle}}{f(x)}$ is the polar function of f .

Proof. Let f be an even log-concave function and consider the function f_m defined as follows

$$f_m(x) = \left(1 + \frac{\log f(x)}{m} \right)_+^m.$$

First, notice that the log-concavity of f implies the $\frac{1}{m}$ -concavity of f_m . Since for all x , $e^{-x} \geq 1 - x$, we get

$$\left(1 - \frac{1}{m} \langle x, y \rangle \right)_+^m \leq e^{-\langle x, y \rangle}.$$

Thus, $\mathcal{L}_{\frac{1}{m}} f \leq \mathcal{L}_0 f = f^\circ$. In addition, it is easy to see that $f_m \leq f$ for any $m > 0$, and since a log-concave function is continuous on its support one has $\lim_{m \rightarrow \infty} f_m = f$ locally uniformly on

$\mathbb{R}^2$. Hence, we can conclude that $\lim_{m \rightarrow \infty} \int_{\mathbb{R}^2} f_m = \int_{\mathbb{R}^2} f$. The fact that $\mathcal{L}_{\frac{1}{m}} f \leq f^\circ$ implies that $\mathcal{L}_{\frac{1}{m}} f_m \leq f_m^\circ$. Moreover, one has

$$\frac{16m^2}{(m+1)(m+2)} \leq \int_{\mathbb{R}^2} f_m \int_{\mathbb{R}^2} \mathcal{L}_{\frac{1}{m}} f_m \leq \int_{\mathbb{R}^2} f_m \int_{\mathbb{R}^2} f_m^\circ.$$

On the other hand, since the functions f_m are log-concave and $\lim_{m \rightarrow \infty} f_m = f$, then using lemma 3.2 in [AKM], we know that $\lim_{m \rightarrow \infty} f_m^\circ = f^\circ$ locally uniformly on the interior of the support of f° . Applying the same lemma again on the log-concave function f_m° , we get that $\lim_{m \rightarrow \infty} \int_{\mathbb{R}^2} f_m^\circ = \int_{\mathbb{R}^2} f^\circ$. Thus, we conclude that

$$\int_{\mathbb{R}^2} f \int_{\mathbb{R}^2} f^\circ = \lim_{m \rightarrow \infty} \int_{\mathbb{R}^2} f_m \int_{\mathbb{R}^2} f_m^\circ \geq 16.$$

□

Remark 3.4. Notice that we don't get the equality case in this way. Recall that, in [FN], it was proved that there is equality in (3.5) if and only if there exists two Hanner polytopes $K_1 \subset F_1$ and $K_2 \subset F_2$, where F_1 and F_2 are two supplementary subspaces in $\mathbb{R}^2$, with $0 \leq \dim(F_i) \leq 2$, such that for all $(x_1, x_2) \in F_1 \times F_2$

$$f(x_1, x_2) = e^{-\|x_1\|_{K_1}} \mathbb{1}_{K_2}(x_2).$$

It would be natural to conjecture that a concave function f on its support satisfies the equality case in (3.4) if and only if

$$f(x_1, x_2) = (1 - \|x_1\|_{K_1})_+ \mathbb{1}_{K_2}(x_2),$$

with K_1, K_2, F_1, F_2 as before. It is easy to see that such functions satisfy the equality case. Nevertheless, clearly, the function $f(x_1, x_2) = (1 - |x_1|)_+ \mathbb{1}_{B_1^2}(x_1, x_2)$ satisfies also the equality case but cannot be written in this form.

For the proof of theorem 3.2, let us start by proving that every even concave function $f : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ can be m -equipartitioned by the canonical basis of $\mathbb{R}^2$.

Lemma 3.5. *Let $m > 0$ and $f : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be an even concave function such that $0 < \int_{\mathbb{R}^2} f < +\infty$. Then there exists a linear invertible map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that the function $f \circ T$ is m -equipartitioned.*

Proof. The proof follows the proof of Lemma 4.2 in our previous work [FN] but we reproduce it for completeness. For any $u \in S^1$, let $C(u) \subset S^1$ be the open half-circle delimited by u and $-u$ containing the vectors v which are after u with respect to the counterclockwise orientation of S^1 . For $v \in C(u)$, let $C(u, v) = \mathbb{R}_+ u + \mathbb{R}_+ v$ be the cone generated by u and v and define

$$f_u(v) = \int_{C(u, v)} f(x)^m dx.$$

The map f_u is continuous and increasing on $C(u)$, $f_u(u) = 0$ and, since f is even, $f_u(v) \rightarrow \frac{1}{2} \int_{\mathbb{R}^2} f^m(x) dx$ when $v \rightarrow -u$. Thus, there exists a unique $v(u) \in C(u)$ such that

$$f_u(v(u)) = \int_{C(u, v(u))} f^m(x) dx = \frac{1}{4} \int_{\mathbb{R}^2} f^m(x) dx.$$

Notice that $v : S^1 \rightarrow S^1$ is continuous and, since f is even, one has

$$\int_{C(v(u), -u)} f^m(x) dx = \frac{1}{4} \int_{\mathbb{R}^2} f^m(x) dx,$$

thus $v(v(u)) = -u$ for any $u \in S^1$. For $u \in S^1$, let

$$g(u) = \int_{C(u, v(u))} f^{m-1}(x) dx - \frac{1}{4} \int_{\mathbb{R}^2} f^{m-1}(x) dx.$$

Then, g is continuous on S^1 and, since f is even,

$$\begin{aligned}
g(v(u)) &= \int_{C(v(u), -u)} f^{m-1}(x) dx - \frac{1}{4} \int_{\mathbb{R}^2} f^{m-1}(x) dx \\
&= \frac{1}{2} \int_{\mathbb{R}^2} f^{m-1}(x) dx - \int_{C(u, v(u))} f^{m-1}(x) dx - \frac{1}{4} \int_{\mathbb{R}^2} f^{m-1}(x) dx \\
&= \frac{1}{4} \int_{\mathbb{R}^2} f^{m-1}(x) dx - \int_{C(u, v(u))} f^{m-1}(x) dx \\
&= -g(u).
\end{aligned}$$

Hence, by the intermediate value theorem, there exists $u \in S^1$ such that $g(u) = 0$, thus

$$\int_{C(u, v(u))} f^{m-1} = \frac{1}{4} \int_{\mathbb{R}^2} f^{m-1} \quad \text{and} \quad \int_{C(u, v(u))} f^m = \frac{1}{4} \int_{\mathbb{R}^2} f^m.$$

□

Let us proceed to the proof of our main theorem.

Proof of Theorem 3.2. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ be an even concave function such that $0 < \int_{\mathbb{R}^2} f < +\infty$. Let $L = B_\infty^m$. For simplicity, the convex body $K_{B_\infty^m, m}(f)$ introduced in section 2 will be denoted in this case by $K_{\infty, m}(f)$ and its dual $K_{B_1^m, m}(\mathcal{L}f)$ will be denoted by $K_{1, m}(\mathcal{L}f)$. Thus, we know that

$$P(K_{\infty, m}(f)) = P(B_\infty^m) P_m(f) = \frac{4^m}{m!} P_m(f).$$

The proof of the theorem concludes if $K_{\infty, m}$ verifies Mahler's conjecture in dimension $m+2$, i.e. $P(K_{\infty, m}) \geq \frac{4^{m+2}}{(m+2)!}$. For that, it is enough to prove that the convex body $K_{\infty, m}$ verifies the hypotheses of Proposition 1.11. We proceed by induction on $m \in \mathbb{N}$. Set, for any positive integer k and for any $\varepsilon \in \{-1, 1\}^k$, $\mathbb{R}_\varepsilon^k = \{x \in \mathbb{R}^k; \varepsilon_i x_i \geq 0, \forall i \in \{1, \dots, k\}\}$.

- Let us start by proving that $K_{\infty, m}(f)$ can be partitioned into 2^{m+2} pieces of the same volume. Since f is m -equipartitioned, we get

$$\begin{aligned}
|K_{\infty, m}(f) \cap \mathbb{R}_+^{m+2}| &= |\{(x, t) \in \mathbb{R}_+^2 \times \mathbb{R}_+^m; \|t\|_\infty \leq f(x)\}| = \int_{\mathbb{R}_+^2} f(x)^m dx |B_\infty^m \cap \mathbb{R}_+^m| \\
&= \int_{\mathbb{R}_+^2} f(x)^m dx = \frac{1}{2^2 |B_\infty^m|} \int_{\mathbb{R}^2} f(x)^m |B_\infty^m| dx = \frac{1}{2^{m+2}} |K_{\infty, m}(f)|.
\end{aligned}$$

Using the fact that $\int_{\mathbb{R}_+^2} f(x)^m dx = \int_{\mathbb{R}_+ \times \mathbb{R}_-} f(x)^m dx$, we get,

$$|K_{\infty, m}(f) \cap \mathbb{R}_\varepsilon^{m+2}| = \frac{1}{2^{m+2}} |K_{\infty, m}(f)| \quad \forall \varepsilon \in \{-1, 1\}^{m+2}.$$

- Our next step is to prove that $K_{\infty, m}(f) \cap e_i^\perp$ verifies Mahler's conjecture for all $i \in \{1, \dots, m+2\}$.

For $i = 1$ or $i = 2$, the same argument holds so let us assume that $i = 1$. One has

$$K_{\infty, m}(f) \cap e_1^\perp = \{(0, x_2, t) \in \mathbb{R}^2 \times \mathbb{R}^m; \|t\|_\infty \leq f(0, x_2)\}.$$

Thus,

$$|K_{\infty, m}(f) \cap e_1^\perp| = 2^m \int_{\mathbb{R}} f(0, x_2)^m dx_2.$$

Since, the dual of a section is the projection of the dual, one has

$$\left(K_{\infty, m}(f) \cap e_1^\perp\right)^\circ = \left(K_{\infty, m}\left(f|_{e_1^\perp}\right)\right)^\circ = K_{1, m}\left(\mathcal{L}\left(f|_{e_1^\perp}\right)\right).$$

Thus

$$\left| \left(K_{\infty,m}(f) \cap e_1^\perp \right)^\circ \right| = \frac{2^m}{m!} \int_{\mathbb{R}} \left(\mathcal{L}(f|_{e_1^\perp})(y) \right)^m dy.$$

Our Theorem 3.1 in dimension 1 implies that

$$\int_{\mathbb{R}} (f(0, x_2))^m dx_2 \int_{\mathbb{R}} (\mathcal{L}(f|_{e_1^\perp})(y))^m dy \geq \frac{4}{m+1}.$$

Hence,

$$P(K_{\infty,m}(f) \cap e_1^\perp) = \frac{4^m}{m!} \int_{\mathbb{R}} (f(0, x_2))^m dx_2 \int_{\mathbb{R}} (\mathcal{L}(f|_{e_1^\perp})(y))^m dy \geq \frac{4^{m+1}}{(m+1)!}.$$

which implies that $K_{\infty,m}(f) \cap e_1^\perp$ verifies Mahler's conjecture in $\mathbb{R}^{m+1}$.

Now, for $i \geq 3$, again, by symmetry, we may assume that $i = 3$. One has

$$K_{\infty,m}(f) \cap e_3^\perp = \{(x, t) \in \mathbb{R}^2 \times (\mathbb{R}^m \cap e_3^\perp); \|t\|_\infty \leq f(x)\} = K_{\infty,m-1}(f).$$

Hence,

$$\begin{aligned} P(K_{\infty,m}(f) \cap e_3^\perp) &= P(K_{\infty,m-1}(f)) = \frac{4^{m-1}}{(m-1)!} \int_{\mathbb{R}^2} f(x)^{m-1} dx \int_{\mathbb{R}^2} (\mathcal{L}f(y))^{m-1} dy \\ &\geq \frac{4^{m-1}}{(m-1)!} \frac{16}{m(m+1)} = \frac{4^{m+1}}{(m+1)!}. \end{aligned}$$

where the last inequality occurs by the induction hypothesis.

- Finally, let us show that $K_{\infty,m}(f) \cap e_i^\perp$, for any $i \in \{1, \dots, m+1\}$, can be partitioned into 2^{m+1} pieces of the same $(m+1)$ -volume. For $i = 1$ (and the same follows for $i = 2$), since f is even, one has

$$\begin{aligned} \left| K_{\infty,m}(f) \cap e_1^\perp \cap \mathbb{R}_+^{m+1} \right|_{m+1} &= \left| \{(0, x_2, t) \in \mathbb{R}_+^2 \times \mathbb{R}_+^m; \|t\|_\infty \leq f(0, x_2)\} \right|_{m+1} \\ &= \int_0^{+\infty} |f(0, x_2) B_\infty^m \cap \mathbb{R}_+^m|_m dx_2 \\ &= \int_0^\infty f(0, x_2)^m dx_2 \\ &= \frac{1}{2^{m+1}} \left| K_{\infty,m}(f) \cap e_1^\perp \right|_{m+1}. \end{aligned}$$

For $i = 3$ (and the same follows for $i > 3$),

$$K_{\infty,m}(f) \cap e_3^\perp \cap \mathbb{R}_+^{m+1} = \{(x, t) \in \mathbb{R}_+^2 \times (\mathbb{R}_+^m \cap e_1^\perp); \|t\|_\infty \leq f(x)\} = K_{\infty,m-1}(f) \cap \mathbb{R}_+^{m+1}.$$

Thus,

$$\left| K_{\infty,m}(f) \cap e_3^\perp \cap \mathbb{R}_+^{m+1} \right|_{m+1} = \int_{\mathbb{R}_+^2} f(x)^{m-1} dx \times |B_\infty^{m-1} \cap \mathbb{R}_+^{m-1}|_{m-1} = \int_{\mathbb{R}_+^2} f(x)^{m-1} dx,$$

and the result follows from the m -equipartition condition. □

Remark 3.6. Notice that if one replaces in the above argument the cube B_∞^m by any unconditional convex body L in $\mathbb{R}^m$, the same proof shows that $K_{L,m}(f)$ satisfies Mahler's conjecture in dimension $m+2$.

4. PROOF OF THE INEQUALITIES IN DIMENSION 1 AND 2 FOR $-\frac{1}{n} < s < 0$.

Let $-\frac{1}{n} < s = -\frac{1}{m} < 0$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}_+$ be an even s -concave function. Then, the function $f = g^s : \mathbb{R}^n \rightarrow (0, +\infty]$ is even and convex. In this section, we give a sharp lower bound of the operator Q_m applied to even convex functions f in dimensions 1 and 2. First, we start by proving the following theorem, which is the analogue version of Theorem 1.6 for convex functions rather than s -concave ones, as seen in equation (2.4).

Theorem 4.1. *Let $f : \mathbb{R}^2 \rightarrow (0, +\infty)$ be an even convex function such that $f \in \mathcal{F}$. Then, for any integer $m > 2$, one has*

$$\int_{\mathbb{R}^2} \frac{1}{(f(x))^m} dx \int_{\mathbb{R}^2} \frac{1}{(\mathcal{M}f(y))^m} dy \geq \frac{16m}{(m-1)(m-2)^2}.$$

Moreover, there is equality for $f(x) = \max(1, \|x\|_\infty)$.

Proof. Let us recall equation (2.6) and apply it in dimension n , which gives that, for all $m > n$ and for every symmetric convex body $K \in \mathbb{R}^{m-n}$, one has

$$\int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy = \left(\frac{m}{m-n} \right)^2 \frac{1}{|K||K^\circ|} |C_{m-n}^K(f)| |C_{m-n}^K(f)^\circ|,$$

where $C_{m-n}^K(f) = \left\{ (x, s) \in \mathbb{R}^n \times \mathbb{R}^{m-n}; \|s\|_K f\left(\frac{x}{\|s\|_K}\right) \leq 1 \right\}$. Applying it for $K = B_\infty^{m-n}$, we simplify the notation of the associated convex body to

$$C_{m-n}^\infty(f) = \left\{ (x, s) \in \mathbb{R}^n \times \mathbb{R}^{m-n}; \|s\|_\infty f\left(\frac{x}{\|s\|_\infty}\right) \leq 1 \right\}.$$

And its dual is denoted by $C_{m-n}^1(\mathcal{M}f)$. In addition, if $C_{m-n}^\infty(f)$ verifies Mahler's inequality in dimension m , we get

$$\begin{aligned} \int_{\mathbb{R}^n} \frac{1}{f(x)^m} dx \int_{\mathbb{R}^n} \frac{1}{(\mathcal{M}f(y))^m} dy &= \left(\frac{m}{m-n} \right)^2 \frac{1}{|B_\infty^{m-n}||B_1^{m-n}|} |C_{m-n}^\infty(f)| |C_{m-n}^\infty(f)^\circ| \\ &\geq \left(\frac{m}{m-n} \right)^2 \frac{(m-n)! 4^m}{4^{m-n} m!} \\ &= \frac{m}{m-n} \times \frac{4^n}{(m-n) \cdots (m-1)}. \end{aligned}$$

Thus, we get the desired inequality (2.4) in dimension n . Since the m -equipartition condition, which is the key of our proof, is only verified in dimension 2, it remains to prove that $C_{m-2}^\infty(f)$ verifies the hypothesis of Proposition 1.11 in dimension m for any integer $m \geq 3$.

We prove it by induction on m . For $m = 3$, $C_{m-2}^\infty(f) = C_1^\infty(f)$ is a symmetric convex body in $\mathbb{R}^3$ hence it satisfies Mahler's conjecture by [IS].

Let $m \geq 4$ and assume that Mahler's conjecture is satisfied for $C_{m-3}^\infty(f)$ and let us prove it for $C_{m-2}^\infty(f)$. For that, recall that $\mathbb{R}_\varepsilon^2 = \{x \in \mathbb{R}^2; \varepsilon_i x_i \geq 0, \forall i \in \{1, 2\}\}$, $\forall \varepsilon \in \{-1, 1\}^2$. We can assume that the even convex function $f : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ is m -equipartitioned. In the sense that, $\forall \varepsilon \in \{-1, 1\}^2$, one has

$$\int_{\mathbb{R}_\varepsilon^2} \frac{dx}{f(x)^m} = \frac{1}{4} \int_{\mathbb{R}^2} \frac{dx}{f(x)^m} \quad \text{and} \quad \int_{\mathbb{R}_\varepsilon^2} \frac{dx}{f(x)^{m-1}} = \frac{1}{4} \int_{\mathbb{R}^2} \frac{dx}{f(x)^{m-1}}.$$

We use proposition 1.11 by proving that the convex body $C_{m-2}^\infty(f)$ verifies its conditions to deduce the lower bound of Mahler's volume of $C_{m-2}^\infty(f)$.

- $C_{m-2}^\infty(f)$ can be partitioned into 2^m pieces of the same volume.

$$C_{m-2}^\infty(f) \cap \mathbb{R}_+^m = \left\{ (x, s) \in \mathbb{R}_+^2 \times \mathbb{R}_+^{m-2}; \|s\|_\infty f\left(\frac{x}{\|s\|_\infty}\right) \leq 1 \right\}.$$

Then,

$$|C_{m-2}^\infty(f) \cap \mathbb{R}_+^m| = \frac{m-2}{m} |B_\infty^{m-2} \cap \mathbb{R}_+^2| \int_{\mathbb{R}_+^2} \frac{dx}{f(x)^m} = \frac{m-2}{4m} \int_{\mathbb{R}^2} \frac{dx}{f(x)^m}.$$

Thus,

$$|C_{m-2}^\infty(f)| = \frac{m-2}{m} |B_\infty^{m-2}| \int_{\mathbb{R}^2} \frac{dx}{f(x)^m} = \frac{m-2}{m} 2^{m-2} \int_{\mathbb{R}^2} \frac{dx}{f(x)^m} = 2^m |C_{m-2}^\infty(f) \cap \mathbb{R}_+^m|.$$

Similarly, since f is even and m -equipartitioned, we can easily prove that for all $\varepsilon \in \{-1, 1\}^m$

$$|C_{m-2}^\infty(f) \cap \mathbb{R}_\varepsilon^m| = \frac{1}{2^m} |C_{m-2}^\infty(f)|.$$

- $C_{m-2}^\infty(f) \cap e_i^\perp$ verifies Mahler in dimension $m-1$, for $1 \leq i \leq m$.

For $i = 1$ (the same holds for $i = 2$),

$$C_{m-2}^\infty(f) \cap e_1^\perp = \left\{ (0, x_2, s) \in \mathbb{R}^2 \times \mathbb{R}^{m-2}; \|s\|_\infty f\left(\frac{(0, x_2)}{\|s\|_\infty}\right) \leq 1 \right\} = \{0\} \times C_{m-2}^\infty(f|_{e_1^\perp}).$$

One has,

$$\left(C_{m-2}^\infty(f) \cap e_1^\perp\right)^\circ = \left(\{0\} \times C_{m-2}^\infty(f|_{e_1^\perp})\right)^\circ = \{0\} \times \left(C_{m-2}^\infty(f|_{e_1^\perp})\right)^\circ = \{0\} \times C_{m-2}^1(\mathcal{M}(f|_{e_1^\perp})).$$

Thus, we get the following volumes

$$\left|C_{m-2}^\infty(f) \cap e_1^\perp\right| = \left|C_{m-2}^\infty(f|_{e_1^\perp})\right| = \frac{m-2}{m-1} 2^{m-2} \int_{\mathbb{R}} \frac{dx_2}{f|_{e_1^\perp}(x_2)^{m-1}}.$$

And

$$\left|\left(C_{m-2}^\infty(f) \cap e_1^\perp\right)^\circ\right| = \left|C_{m-2}^1(\mathcal{M}(f|_{e_1^\perp}))\right| = \frac{(m-2)2^{m-2}}{(m-1)(m-2)!} \int_{\mathbb{R}} \frac{dx_2}{(\mathcal{M}f|_{e_1^\perp}(x_2))^{m-1}}.$$

Applying the result in dimension 1, proved in [FGSZ], to $f|_{e_1^\perp}$ we get

$$\begin{aligned} \left|C_{m-2}^\infty(f) \cap e_1^\perp\right| \left|\left(C_{m-2}^\infty(f) \cap e_1^\perp\right)^\circ\right| &= \frac{(m-2)^2 4^{m-2}}{(m-1)^2 (m-2)!} \int_{\mathbb{R}} \frac{dx_2}{f|_{e_1^\perp}(x_2)^{m-1}} \int_{\mathbb{R}} \frac{dx_2}{(\mathcal{M}f|_{e_1^\perp}(x_2))^{m-1}} \\ &\geq \frac{(m-2)^2 4^{m-2}}{(m-1)^2 (m-2)!} \frac{m-1}{m-2} \frac{4}{m-2} = \frac{4^{m-1}}{(m-1)!}. \end{aligned}$$

For $i = 3$ and the case $i > 3$ follows similarly:

$$C_{m-2}^\infty(f) \cap e_3^\perp = \left\{ (x, 0, s) \in \mathbb{R}^2 \times \{0\} \times \mathbb{R}^{m-3}; \|(0, s)\|_\infty f\left(\frac{x}{\|(0, s)\|_\infty}\right) \leq 1 \right\} = C_{m-3}^\infty(f).$$

Thus,

$$\left|C_{m-2}^\infty(f) \cap e_3^\perp\right| = |C_{m-3}^\infty(f)| = \frac{m-3}{m-1} 2^{m-3} \int_{\mathbb{R}^2} \frac{dx}{f(x)^{m-1}}.$$

And

$$\left|\left(C_{m-2}^\infty(f) \cap e_3^\perp\right)^\circ\right| = \left|(C_{m-3}^\infty(f))^\circ\right| = |C_{m-3}^1(\mathcal{M}f)| = \frac{m-3}{m-1} \frac{2^{m-3}}{(m-3)!} \int_{\mathbb{R}^2} \frac{dy}{(\mathcal{M}f(y))^{m-1}}.$$

Then, using the hypothesis of induction on m , we obtain,

$$\begin{aligned} \left| C_{m-2}^\infty(f) \cap e_3^\perp \right| \left| \left(C_{m-2}^\infty(f) \cap e_3^\perp \right)^\circ \right| &= \frac{(m-3)^2}{(m-1)^2} \frac{4^{m-3}}{(m-3)!} \int_{\mathbb{R}^2} \frac{dx}{f(x)^{m-1}} \int_{\mathbb{R}^2} \frac{dy}{(\mathcal{M}f(y))^{m-1}} \\ &\geq \frac{4^{m-1}}{(m-1)!}. \end{aligned}$$

which concludes the second part.

- $C_{m-2}^\infty(f) \cap e_i^\perp$ can be partitioned into 2^{m-1} pieces of the same volume, for $1 \leq i \leq m$.

For $i = 1$ and $i = 2$, one has

$$C_{m-2}^\infty(f) \cap e_1^\perp = \{0\} \times C_{m-2}^\infty(f|_{e_1^\perp}) \quad \text{and} \quad C_{m-2}^\infty(f) \cap e_2^\perp = C_{m-2}^\infty(f|_{e_2^\perp}) \times \{0\}.$$

These convex bodies are unconditional, thus we get the partition.

For $i = 3$, by the m -equipartition condition of f , we obtain

$$\begin{aligned} \left| C_{m-2}^\infty(f) \cap e_3^\perp \cap \mathbb{R}_+^{m-1} \right| &= \left| C_{m-3}^\infty(f) \cap \mathbb{R}_+^{m-1} \right| = \frac{m-3}{m-1} \left| B_\infty^{m-3} \cap \mathbb{R}_+^{m-1} \right| \int_{\mathbb{R}_+^2} \frac{dx}{f(x)^{m-1}} \\ &= \frac{1}{4} \frac{m-3}{m-1} \int_{\mathbb{R}^2} \frac{dx}{f(x)^{m-1}} = \frac{1}{2^{m-1}} \left| C_{m-2}^\infty(f) \cap e_3^\perp \right|. \end{aligned}$$

Using the same method, one may prove the partition of the body $C_{m-2}^\infty(f) \cap e_i^\perp$ for $i > 3$.

Therefore $C_{m-2}^\infty(f)$ satisfies Mahler's conjecture, which concludes the proof of the inequality of Theorem 4.1. Moreover, it is easy to check that there is equality for $f(x) = \max(1, \|x\|_\infty)$, since, in this case $C_{m-2}^\infty(f) = B_\infty^m$. $\square$

Our next goal in this section is to prove the following theorem in dimension 1, which establishes Mahler's conjecture in dimension 1 for s -concave functions for any $-1 < s < 0$.

Theorem 4.2. *Let $f : \mathbb{R} \rightarrow (0, +\infty]$ be an even convex function such that $\lim_{t \rightarrow +\infty} f(tx) = +\infty$, for any $x \neq 0$. Then, for any $m > 1$, one has*

$$Q_m(f) = \int_{\mathbb{R}} \frac{dx}{f(x)^m} \int_{\mathbb{R}} \frac{dy}{(\mathcal{M}f(y))^m} \geq Q_m(1 + I_{[-1,1]}) = \frac{4}{m-1},$$

with equality if and only if $f = c + I_{[-\alpha, \alpha]}$, for some $c, \alpha > 0$.

We first show that Theorem 4.2 follows from the next theorem, which implies Theorem 1.8, if one applies it to $f = g^s$, for any g , s -concave, for $-1 < s < 0$ and $m = -1/s$.

Theorem 4.3. *Let $f : \mathbb{R} \rightarrow (0, +\infty]$ be an even convex function such that $\lim_{t \rightarrow +\infty} f(tx) = +\infty$, for any $x \neq 0$. Then for any $m > 1$,*

$$\int_{\mathbb{R}} \frac{1}{f(x)^m} dx \geq \frac{m-1}{m} \int_{\mathbb{R}} \frac{1}{(\mathcal{M}f(x))^m} dx, \quad (4.1)$$

with equality if and only if $f = c + I_{[-\alpha, \alpha]}$, for some $c, \alpha > 0$.

Multiplying equation (4.1) by $\int (\mathcal{M}f)^{-m}$ we deduce that for any even convex f tending to $+\infty$ at infinity, one has

$$Q_m(f) \geq \frac{m-1}{m} Q_m(\mathcal{M}f), \quad (4.2)$$

with equality if and only if $f = c + I_{[-\alpha, \alpha]}$, for some $c, \alpha > 0$. Since $\mathcal{M}f \in \mathcal{F}$, using [FGSZ] we get

$$Q_m(\mathcal{M}f) \geq Q_m(1 + |x|) = \frac{4m}{(m-1)^2}. \quad (4.3)$$

Using equations (4.2) and (4.3), we conclude the proof of Theorem 4.2.

Proof of Theorem 4.3. Let us define a variant of the convex body C_m^K introduced in section 2, which is more suitable for $m = 1/|s| \notin \mathbb{N}$, this variant was considered in [IW] for $s > 0$:

$$C_1(f) = C_1^{[-1,1]}(f) = \{(x, t) \in \mathbb{R} \times \mathbb{R}; \pi_1(f)(x, t) \leq 1\} = \left\{ (x, t) \in \mathbb{R} \times \mathbb{R}; |t|f\left(\frac{x}{|t|}\right) \leq 1 \right\}.$$

The same computations as before show easily that $C_1(f)^\circ = C_1(\mathcal{M}f)$. Moreover, for any $m > 1$, one has

$$\int_{\mathbb{R}} \frac{1}{f(x)^m} dx = \int_{\mathbb{R}} \int_{f(x)}^{+\infty} \frac{m}{t^{m+1}} dt dx = \int_0^{+\infty} \frac{m}{t^{m+1}} |\{x \in \mathbb{R}; f(x) \leq t\}| dt.$$

Using the change of variables $t = 1/u$, we get

$$\int_{\mathbb{R}} \frac{1}{f(x)^m} dx = \frac{m}{2} \int_{C_1(f)} |u|^{m-2} du dx = \mu_m(C_1(f)),$$

where we denote by μ_m the measure with density $d\mu_m(x, t) = \frac{m}{2}|t|^{m-2} dx dt$ on $\mathbb{R}^2$. Applied to $\mathcal{M}\mathcal{M}f$ instead of f , we get

$$\int_{\mathbb{R}} \frac{dz}{(\mathcal{M}\mathcal{M}f(z))^m} = \frac{m}{2} \int_{C_1(\mathcal{M}\mathcal{M}f)} |s|^{m-2} ds dx = \mu_m(C_1(\mathcal{M}\mathcal{M}f)).$$

Since $C_1(\mathcal{M}\mathcal{M}f) = C_1(\mathcal{M}f)^\circ = C_1(f)^{\circ\circ} = \text{Conv}(C_1(f))$, we get

$$\int_{\mathbb{R}} \frac{dz}{(\mathcal{M}\mathcal{M}f(z))^m} = \frac{m}{2} \int_{\text{Conv}(C_1(f))} |s|^{m-2} ds dx = \mu_m(\text{Conv}(C_1(f))).$$

The inequality (4.1) that we want to prove is thus equivalent to

$$\mu_m(C_1(f)) \geq \frac{m-1}{m} \mu_m(\text{Conv}(C_1(f))). \quad (4.4)$$

It was proved in [FGSZ, Theorem 2.16] that $C_1(f)$ can be written as $C_1(f) = C_1(f)_+ \cup C_1(f)_-$, where

$$C_1(f)_+ = C_1(f) \cap \{s \geq 0\} = \overline{\left\{ (x, s) \in \mathbb{R} \times (0, +\infty); sf\left(\frac{x}{s}\right) \leq 1 \right\}}$$

is a convex body containing 0 on its boundary and $C_1(f)_-$ is its symmetric image with respect to the hyperplane $\{s = 0\}$. Thus, our problem can be formulated as follows:

for every convex body $K \subset \{(x, s) \in \mathbb{R}^2; s \geq 0\}$ such that $0 \in \partial K$ and K symmetric with respect to $\mathbb{R}e_2$ and if we denote $C = K \cup \sigma(K)$, where $\sigma = \sigma_{e_2^\perp}$ then

$$\mu_m(C) \geq \frac{m-1}{m} \mu_m(\text{Conv}(C)). \quad (4.5)$$

Let $a \in \mathbb{R}$ such that $P_{e_2^\perp}(K) = [-a, a]$, then there exists $b \geq 0$ such that $(a, b) \in K$. Set

$$S = K \cap \{s = b\} = [-a, a] \times \{b\}, \quad K_1 = \text{Conv}(S, 0), \quad \text{and} \quad C_1 = K_1 \cup \sigma(K_1).$$

Then, one has $C_1 \subset C$ and one easily sees that $\text{Conv}(C) \setminus C \subset \text{Conv}(C_1) \setminus C_1$, thus

$$\frac{\mu_m(\text{Conv}(C) \setminus C)}{\mu_m(C)} \leq \frac{\mu_m(\text{Conv}(C_1) \setminus C_1)}{\mu_m(C_1)}. \quad (4.6)$$

The triangle K_1 with vertices 0, (a, b) and $(-a, b)$ can be described as

$$K_1 = \{(x, s) \in \mathbb{R}^2; \frac{b}{a}|x| \leq s \leq b\}.$$

Hence, we get

$$\mu_m(C_1) = 2\mu_m(K_1) = 4 \int_0^a \int_{\frac{b}{a}x}^b s^{m-2} ds dx = \frac{4}{m} ab^{m-1}.$$

And since $\text{Conv}(C_1) = [-a, a] \times [-b, b]$, we get

$$\mu_m(\text{Conv}(C_1)) = 4 \int_0^a \int_0^b s^{m-2} ds dx = \frac{4}{m-1} ab^{m-1}.$$

Hence

$$\frac{\mu_m(\text{Conv}(C_1) \setminus C_1)}{\mu_m(C_1)} = \frac{\frac{1}{m-1} - \frac{1}{m}}{\frac{1}{m}} = \frac{1}{m-1}.$$

Using (4.6), we deduce that (4.5) holds. Moreover, there is equality in (4.6) if and only if $C = C_1$, i.e. $K = K_1$, which means that $f = \frac{1}{b} + I_{[-\frac{a}{b}, \frac{a}{b}]}$. Thus, the equality case follows. $\square$

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